



Lithium Difluoro(oxalato)borate as Salt for Lithium-Ion Batteries

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Lithium difluoro(oxalato)borate is reported as a salt for high-performance lithium-ion batteries with improved cycle life and power capability. The experimental results showed that lithium difluoro(oxalato)borate, $\text{LiC}_2\text{O}_4\text{BF}_2$, can be reduced at about 1.7 V vs Li^+/Li and forms a robust protective SEI film on the graphite surface, as lithium bis(oxalato)borate does. The lithium-ion cells using lithium difluoro(oxalato)borate-based electrolyte had very good capacity retention at 55°C. The lithium-ion cells using the lithium difluoro(oxalato)borate-based electrolyte had very low interfacial impedance. Therefore, lithium difluoro(oxalato)borate is a promising salt for advanced lithium-ion batteries with improved capacity retention and power capability.
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Since the first commercialization of lithium-ion batteries by Sony Corp., lithium hexafluorophosphate (LiPF_6) remains the dominant salt for lithium-ion batteries. A great deal of effort has been devoted to developing alternative lithium salts to replace LiPF_6 , which is known to generate HF acid when exposed to moisture. However, a successful replacement of LiPF_6 has not occurred due to the multiple criteria that a good salt must meet, and because of the complex chemical and electrochemical environment in lithium-ion cells. For example, the thermal reactivity of LiClO_4 is a big concern for the safety of lithium-ion batteries, while the corrosion of aluminum positive current collector in lithium imide electrolytes is a big challenge to overcome.¹⁻³ Recently, lithium bis(oxalato)borate (LiBOB) was proposed as an alternative to LiPF_6 . In general, LiBOB has fairly good solubility in nonaqueous solvents (carbonates) and provides reasonable lithium ion conductivity.⁴⁻⁶ Most favorably, the solid electrolyte interphase (SEI) formed on the carbonaceous negative electrode by the reduction of the LiBOB -based electrolyte can effectively protect the negative electrode from being attacked by other cell components.⁷⁻⁹ Furthermore, this stable SEI film can also improve the thermal stability of the negative electrodes.^{10,11} However, LiBOB -based electrolytes have their own serious drawback that is related to the highly resistive SEI film formed on the negative electrode surface due to the reduction of LiBOB , which limits the power capability of a cell based on this electrolyte.^{12,13}

In order for lithium-ion batteries to be used as a power source for hybrid electric vehicles (HEV), a dramatic improvement on both capacity and power retention is needed to meet the challenging 15-year life requirement. Cells based on LiPF_6 electrolyte have difficulty meeting this requirement. LiPF_6 is sensitive to trace amounts of moisture and can undergo thermal decomposition at elevated temperatures (above 50°C), resulting in by-products such as PF_5 , POF_3 , HF, and LiF. PF_5 and POF_3 can trigger detrimental reactions on the electrode surface. HF is responsible for the dissolution and migration of the transition metals from the positive electrodes.⁷ Furthermore, LiF, which usually deposits on the electrode surface, can result in high interfacial cell impedance that accelerates the capacity and power fade of the cell. Thus, the side reactions of LiPF_6 make it difficult to meet the 15-year battery life requirement for HEV applications. A lithium-ion cell using LiBOB has very good capacity retention, but poor power capability. One possibility is to use a blend of LiPF_6 and LiBOB to capitalize on the advantages of both salts.⁸ In this work, we report a salt, lithium difluoro(oxalato)borate (LiDFOB), which provides a stable SEI film and low cell impedance. The use of LiDFOB as a promising salt for high-power applications is discussed.

Experimental

The salt investigated here is lithium difluoro(oxalato)borate (LiDFOB), whose anion is schematically shown in Fig. 1a. As a comparison, the structure of bis(oxalato)borate anion (BOB^-) is shown in Fig. 1b. Compared with BOB^- , DFOB^- has only one oxalate subunit and the other two σ bonds are formed with fluorine. The replacement of one oxalate subunit with two fluorine atoms significantly improves the solubility of the salt (LiDFOB) in nonaqueous carbonates. Generally, the concentration of LiDFOB in the conventional carbonate solvents, i.e., the mixture of ethylene carbonate (EC) and ethyl methyl carbonate (EMC), can be greater than 1 M, while the average molarity of LiBOB is about 0.7 M. The electrolytes used in this work include 1.2 M LiPF_6 in ethylene carbonate/propylene carbonate/dimethyl carbonate (EC/PC/DMC, 1:1:3 by weight), 0.7 M LiBOB in EC/PC/DMC (1:1:3 by weight), and 1.0 M LiDFOB in EC/PC/DMC (1:1:3 by weight).

The positive electrode material investigated is $\text{Li}_{1.1}[\text{Mn}_{1/3}\text{Ni}_{1/3}\text{Co}_{1/3}]_{0.9}\text{O}_2$ with 10 atom % excess lithium (referred to subsequently here as L333). The electrodes used were specially designed at Argonne National Laboratory for high-power lithium-ion batteries, and were prepared by a battery manufacturer (Quallion LLC). The positive electrode laminate is composed of 84 wt % L333, 8 wt % carbon black (TB5500), and 8 wt % poly(vinylidene fluoride) (PVDF). The negative electrode laminate is composed of 90 wt % mesocarbon microbeads (MCMB, $D_{50} = 10 \mu\text{m}$), 2 wt % carbon fiber, and 8 wt % PVDF.

The performance of MCMB/L333 cells was evaluated using 2032-type coin cells. The cell stack consisted of a negative electrode, a microporous polypropylene separator (Celgard 3501), a positive electrode, and an appropriate amount of nonaqueous electrolyte.

High-power performance of the lithium-ion cells was investigated by the hybrid pulse power characteristic (HPPC) testing method set by the Department of Energy's FreedomCAR Partnership.¹⁴ This test is intended to determine the dynamic power capability of a battery over its usable charge and voltage range using a test profile that incorporates both discharge and regenerative pulses. For example, the cell was first charged at the 1 C rate to

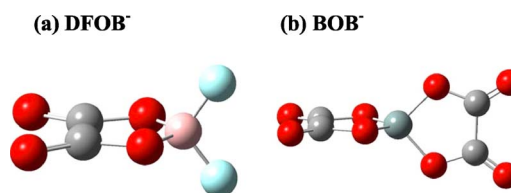


Figure 1. (Color online) Molecular structures of difluoro(oxalato)borate and bis(oxalato)borate anions.

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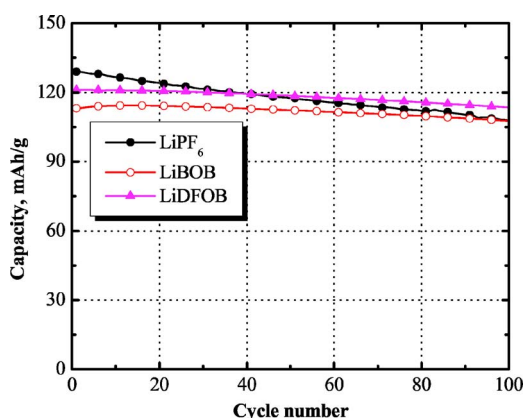


Figure 2. (Color online) Discharge capacity of MCMB/L333 cells cycling between 3.0 and 4.0 V at 55°C with a constant current of C/2.

4.1 V or other top-of-charge specified. After a 1 h rest, the cell was discharged to 10% depth of discharge (DOD) at the 1 C rate and then rested for another hour. This was followed by a series of partial (but increasing) 1 C discharges at 10% DOD increments, each of which was followed by an 18 s pulse discharge and a 2 s pulse charge at a very high rate (5–20 C). The area specific impedance (ASI) was then calculated for both the 18 s pulse discharge and the 2 s pulse charge. The ASI's standard requirement set by FreedomCAR to meet the power requirement for HEV is 35 $\Omega \text{ cm}^2$ for the 18 s pulse discharge and 25 $\Omega \text{ cm}^2$ for the 2 s pulse charge.¹⁴ A detailed description of HPPC test is provided in a recent publication.¹⁵

Results and Discussion

Figure 2 shows the specific capacity of MCMB/L333 lithium-ion cells that were cycled at 55°C with a constant current of C/2 (~1.2 mA). The electrolytes used are 1.2 M LiPF₆, 0.7 M LiBOB, and 1.0 M LiDFOB in EC/PC/DMC (1:1:3, by weight). When the LiPF₆-based electrolyte was used, the cell's capacity faded quickly at 55°C; more than 17% capacity was lost in 100 cycles. When a LiBOB-based electrolyte was used, the initial reversible capacity of the cell was lower than that of the cell using LiPF₆. The difference in initial capacity can be explained by the high-impedance SEI layer formed in the LiBOB-based electrolyte. It has been well discussed that the SEI formed in LiBOB-based electrolyte is very effective in protecting the negative electrode from degradation during cycling;⁸ however, this film usually leads to a very high interfacial impedance of the carbon electrode, which limits the power capability of the cell. Figure 2 clearly shows that the cell based on LiBOB electrolyte has much better capacity retention than that based on LiPF₆ electrolyte, with only 6% capacity loss after 100 cycles.

The solid triangles in Fig. 2 illustrate the capacity retention of an MCMB/L333 cell using LiDFOB-based electrolyte. The initial reversible capacity of the cell lies between that for cell using LiPF₆ and cell using LiBOB. This means that LiDFOB does provide lower interfacial impedance than LiBOB does. Furthermore, the capacity retention in this case is very high, about 94% after 100 cycles at 55°C. In addition, the trend of the capacity fade in the cell with LiDFOB-based electrolyte is very similar to that of cells based on LiBOB electrolyte, except for a slightly higher reversible capacity. Clearly, LiDFOB can be the alternative salt to maintain both high capacity retention and high-power performance in lithium-ion cells. In the following sections, the interfacial phenomenon will be discussed further to illustrate the advantages of LiDFOB compared with LiPF₆ and LiBOB.

Figure 3 shows the differential capacity profile of MCMB/Li cells using different electrolytes. The cells were cycled at 25°C with a constant current of C/10, and only the initial data showing the

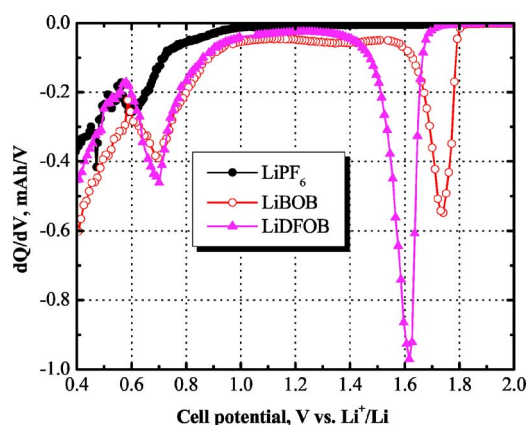


Figure 3. (Color online) Differential capacity profiles of MCMB/Li cells using different electrolytes showing the impact of salt on the SEI formation. The cells were cycled with a constant current of C/10.

formation of SEI are shown in Fig. 3. Only one small peak at about 0.6 V vs Li⁺/Li was observed for the cell using LiPF₆-based electrolyte. This peak can also be observed for cells using LiBOB and LiDFOB at a slightly higher potential. This peak is believed to be the result of the reduction of the solvents during passivation reaction on the graphite surface. However, an extra peak was observed at about 1.7 V vs Li⁺/Li for the cell using LiBOB. This peak (at 1.7 V) is believed to be caused by the reduction and polymerization of BOB⁻ on the surface of graphite, which results in improved capacity retention and high interfacial impedance. Similarly, a strong peak at about 1.6 V was observed for the cell using LiDFOB. Because of the structural similarity between BOB⁻ and DFOB⁻, we believe that the peak at 1.6 V comes from the reduction and polymerization of DFOB⁻, which can also explain the improved capacity retention of the lithium-ion cell using LiDFOB during cycling at high temperature (see Fig. 2).

In order to confirm the impact of the salt on the interfacial impedance of graphite electrodes, three symmetric graphite-graphite cells were prepared. For each electrolyte, two MCMB/Li coin cells were prepared and constant-voltage-discharged to 1 mV before the cells were opened. The two lithiated MCMB electrodes harvested from the half cells were then used to assemble a new symmetric cell using the same electrolyte. After that, the ac impedance of the symmetric cells was measured in the frequency range from 10 mHz to 1 MHz. Figure 4 shows the ac impedance of the symmetric cells using different electrolytes. The data shown in Fig. 4 represent two

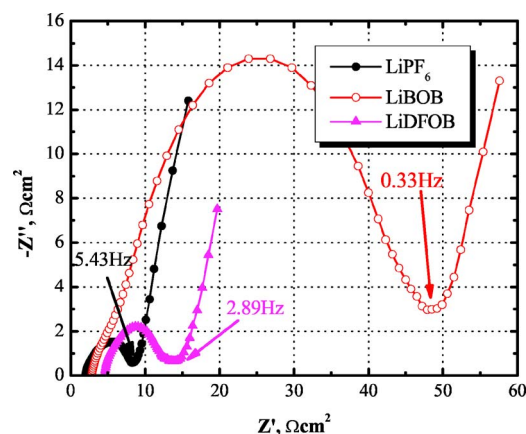


Figure 4. (Color online) AC impedance of lithiated graphite-lithiated graphite cells showing the impact of salt on interfacial impedance.

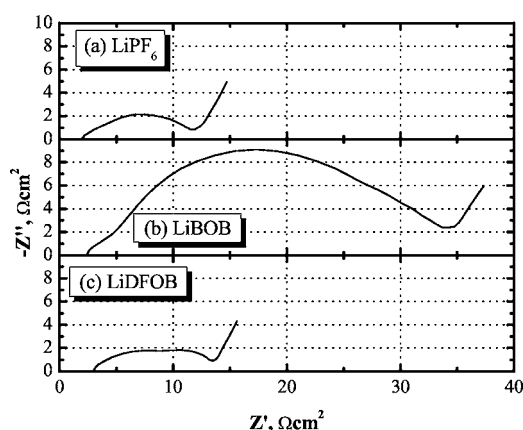


Figure 5. AC impedance of MCMB/L333 cells showing the impact of the salt on cell impedance. The data were collected at a constant voltage of 3.8 V.

layers of SEI, one for each MCMB electrode. The symmetric cell using LiPF_6 has the lowest interfacial impedance, about $9 \Omega \text{ cm}^2$; the cell using LiDFOB-based electrolyte has a slightly higher impedance of about $15 \Omega \text{ cm}^2$, which is much lower than that of the cell using LiBOB-based electrolyte (about $49 \Omega \text{ cm}^2$).

Figure 5 shows the ac impedance of MCMB/L333 cells using different electrolytes. The cells were first cycled between 3.0 and 4.0 V for two cycles with a constant current of C/2. Then, the cells were constant-voltage-charged to 3.8 V before the ac impedance was measured. Apparently, the cell using LiBOB-based electrolyte had very high interfacial impedance, more than double that of the cells using LiPF_6 - and LiDFOB-based electrolytes. From the ac impedance results of both the carbon symmetrical cells and the full cell, it is very clear that the use of LiDFOB provides (i) very stable passivation film that leads to long cycle and calendar life, and (ii) very low interfacial impedance that leads to very high power capability of the cell. The HPPC test was also carried out for the same cells. Figure 6 shows the ASI of the cells as a function of the DOD. When LiPF_6 -based electrolyte was used, the ASI for the 18 s pulse discharge at 10 C rate is about $20 \Omega \text{ cm}^2$, and the ASI for the 2 s regeneration (charge) is about $14 \Omega \text{ cm}^2$. Both data greatly exceed the standard requirement set by FreedomCAR for a battery to power an HEV. When the LiBOB-based electrolyte was used, the cell showed very high ASI of about $33 \Omega \text{ cm}^2$ with a very narrow DOD window. However, when the LiDFOB-based electrolyte was used, the ASI of the cell was similar to that of LiPF_6 -based electrolyte. Cells based on LiDFOB-based electrolyte provide a high-power capability similar to that of LiPF_6 -based cells and excellent cycle life even at elevated temperatures. Therefore, it could be a very promising salt for high-power batteries for HEV applications.

Conclusion

A salt, lithium difluoro(oxalato)borate, was reported as a promising salt for advanced and stable electrolyte that can improve the life of lithium-ion batteries. The unique feature of lithium difluoro(oxalato)borate is that it gives very low interfacial impedance for lithium-ion cells as in the case of LiPF_6 -based electrolyte. Furthermore, the LiDFOB-based electrolyte system provides the same protection of the surface of the negative electrode from degradation as does the LiBOB-based electrolyte with the formation of a stable SEI

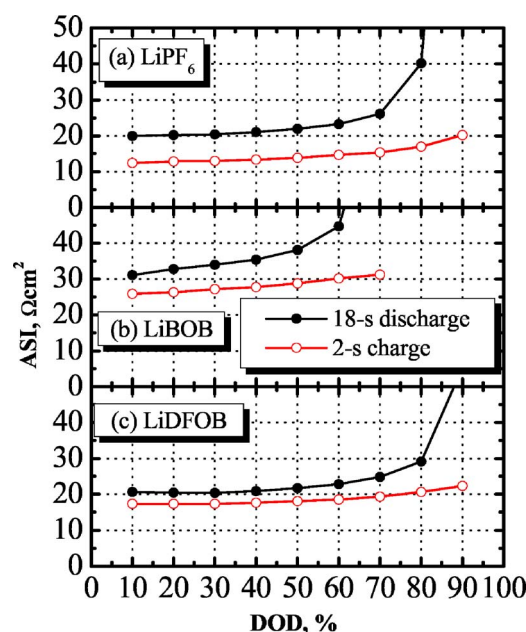


Figure 6. (Color online) ASI of MCMB/L333 cell during HPPC showing the impact of the salt on the power performance of the cells.

film. The MCMB/ $\text{Li}_{1.1}[\text{Mn}_{1/3}\text{Ni}_{1/3}\text{Co}_{1/3}]_{0.9}\text{O}_2$ lithium-ion cell using this salt has excellent capacity retention at 55°C , while its power capability greatly exceeds the standard requirement set by FreedomCAR for applications in hybrid electric vehicles.

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